

CoMDet: A Contrastive Multimodal Pre-Training Approach to Encrypted Malicious Traffic Detection

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Abstract—Encrypted malicious traffic detection aims at identifying malicious activities without decrypting network traffic, which is essential for cybersecurity. Existing methods have shown effective performance in encrypted malicious traffic detection, but their heavy reliance on labeled datasets presents a challenge. This paper presents CoMDet, a contrastive multimodal pre-training approach, to detect encrypted malicious traffic. CoMDet leverages three independent Transformer encoders to learn multimodal feature representations from encrypted traffic based on unlabeled data in the pre-training phase. Meanwhile, we introduce a novel inter-modal contrastive learning method to enhance feature representation by maximizing the mutual information among the modalities. Subsequently, we fine-tune the pre-trained model using limited labeled data. Experimental results demonstrate that CoMDet outperforms the existing semi-supervised learning methods and achieves comparable performance with existing supervised learning methods. It obtains an average *ACC* of 92% and an average *macro-F1* of 86% with only 80 labeled samples in each malicious category. We conduct an investigation on the fine-tuning dataset size and discover that as the dataset size increases, the performance of CoMDet is increasingly comparable to existing supervised learning methods.

Index Terms—Encrypted malicious traffic detection, Contrastive multimodal pre-training, Model fine-tuning, Transformer encoder

I. INTRODUCTION

Nowadays, the extensive use of network traffic encryption plays a crucial role in ensuring the security of data transmission. As of January 2023, encrypted traffic using Hypertext Transfer Protocol Secure (HTTPS) accounted for 94% of all traffic in the products and services offered by Google [1]. Nevertheless, traffic encryption is double-edged, as malicious users may exploit it to conceal illegal activities, including network attacks within encrypted channels (such as scanning, brute-force attacks, and data exfiltration), encrypted traffic generated by malware (such as cryptocurrency mining), and illicit encryption applications, for instance, The Onion Router (TOR) and unauthorized Virtual Private Network (VPN). According to the latest threat research report released by Zscaler [2], attacks based on encrypted channels have shown a continuous upward trend, rising from 57% in 2020 to 85% in 2022. The disparity between encrypted malicious traffic and benign traffic undermines the effectiveness of existing detection methods, since more existing methods are primarily

designed to handle non-encrypted traffic. It is essential to effectively detect encrypted malicious traffic for maintaining cybersecurity.

Methods for detecting encrypted malicious traffic can be roughly divided into two categories: machine learning-based and deep learning-based methods. Most early machine learning-based detection methods often relied on prior knowledge. These methods use plain-text features (such as certificates and protocols) extracted based on expert knowledge to detect encrypted traffic [3]. Once the features of encrypted malicious traffic slightly changed, these methods would fail to detect them. The deep learning-based detection methods [4] learn deep features directly from raw network traffic, enabling them to better handle the constantly evolved encrypted malicious traffic. However, deep learning-based methods heavily rely on labeled datasets with a balanced distribution of different classes. Training deep learning models with inappropriate datasets may lead to overfitting issues and could even result in training failure.

This paper presents **CoMDet**, which is a **Contrastive Multimodal** pre-training approach, to encrypted malicious traffic **Detection**. CoMDet consists of two phases, namely contrastive multimodal pre-training phase and fine-tuning phase. As for the first phase, we pre-train three Transformer encoders to learn general feature representations from unlabeled data. Initially, we transform each traffic session into embeddings using three modalities: SSL/TLS handshake payload bytes, packet length sequences, and packet arrival-time interval sequences. Subsequently, we employ three individual Transformer encoders to encode each modality. Transformer encoders are capable of generating accurate representations of encrypted traffic by maximizing the mutual information among modalities in the same latent space. As for the second phase, CoMDet introduces a new fine-tuning scheme based on limited labeled samples to achieve comparable performance with existing methods.

The main contributions of this paper are summarized as follows:

- We present CoMDet, which is a contrastive multimodal pre-training approach to encrypted malicious traffic detection. CoMDet utilizes pre-trained Transformer encoders to learn general feature representations, enabling fine-tuning with limited labeled samples.

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- CoMDet introduces a novel inter-modal contrastive learning method aimed at maximizing the mutual information among all modalities, resulting in its capability of facilitating enhanced feature representations.
- Experimental results demonstrate that CoMDet achieves notable performance compared with existing semi-supervised learning methods. CoMDet obtains an average *ACC* of 92% and an average *macro* – *F1* of 86% with only 80 labeled samples in each malicious category. Additionally, we conduct a thorough analysis of the impact of CoMDet on the dataset size and malicious traffic categories.

II. RELATED WORK

A. Encrypted Malicious Traffic Detection

Non-encrypted network traffic takes on completely different forms after encryption. It renders existing malicious traffic detection methods, which typically rely on analyzing plaintext features, ineffective. With the advancement of machine learning, researchers adopt various machine learning methods to detect encrypted malicious traffic such as K-Nearest Neighbors (K-NN) [5], Decision Tree (DT) [6], Random Forest (RF) [7]. For instance, MaMPF [8] leverages packet types and packet arrival-time interval sequences to train a Markov model, and achieves 96.4% TPR and 0.2% FPR performance on a real-world dataset from campus network. MEMTD [9] extracts modalities features from encrypted traffic to enhance detection performance, and outperforms other single-modal deep learning methods. Most machine learning-based methods rely on manually-crafted features that require prior domain knowledge to distinguish between malicious and benign traffic [10].

Deep learning-based methods have emerged in recent years. These methods automatically extract deep features from raw network traffic. This enables an end-to-end detection of encrypted malicious traffic. ET-BERT [11] pre-trains deep contextualized datagram-level representations from large-scale unlabeled data, and achieves remarkable performance after fine-tuning with task-specific labeled data. FC-Net [12] enhances performance under conditions with limited samples using meta-learning techniques. It is not confined to specific datasets or attack types. Rezaei *et al.* [13] employ a 1D-CNN (1Dimension-Convolutional Neural Network) model that takes packet arrival-time interval sequences and packet arrival-time interval sequences as input, and achieves almost the same accuracy as supervised methods based on only 20 samples in each class. Although deep learning-based methods outperform machine learning-based methods in detection performance, they heavily rely on labeled data. Collecting and labeling encrypted malicious network traffic in a real-world network environment presents an exceptionally daunting challenge.

B. Contrastive Learning

Contrastive learning has showcased significant advancements in natural language processing and computer vision,

offering a means to automatically learn feature vectors from large-scale unlabeled datasets by discerning similarities between various sample pairs. Contrastive learning methods calculate the loss within the latent space to guide the training process and obtain general representations of samples. To facilitate contrastive learning, He *et al.* [14] introduce a queue structure and a momentum encoder for contrastive learning, outperforming supervised training models in diverse computer vision tasks. Chen *et al.* [15] employ stop-gradient operations to prevent ineffective training, demonstrating the efficacy of contrastive learning in efficiently training high-performance deep learning models.

In the field of encrypted malicious traffic detection, researchers have already explored contrastive learning methods to handle this task [12, 16, 17, 18]. These methods extract robust representation vectors from unlabeled data and enhance the performance of encrypted malicious traffic detection by contrastive learning. Notably, contrastive learning can be implemented through various strategies, which is a flexible and unsupervised representation learning method. Furthermore, it can be extended to inter-modal contrastive learning among multiple modalities by leveraging multimodal datasets [19]. Therefore, leveraging large-scale unlabeled data for contrastive learning methods presents a promising research approach in encrypted malicious traffic detection.

III. CoMDet

In this section, we present CoMDet which includes two phases: (1) contrastive multimodal pre-training, and (2) fine-tuning by limited data.

In the contrastive multimodal pre-training phase, we initialize three individual Transformer encoders to map SSL/TLS handshake payload bytes, packet length sequences, and packet arrival-time interval sequences to the same latent space. We propose an inter-modal contrastive learning method to optimize the feature representation of multiple modalities. It enhances the mutual information among modalities and reduces noise information by using contrastive learning among multimodal feature vectors. As a result, the Transformer encoders learn a better representation that preserves the latent shared information among modalities while removing modality-specific noise.

In the fine-tuning phase, we fine-tune the pre-trained model based on limited labeled data. Additionally, we classify the encrypted traffic categories by a fully connected layer, which receives multimodal features in latent space as input. CoMDet is capable of efficiently fine-tuning with limited data, capitalizing on its well-learned representations. The overview of CoMDet is shown in Fig 1.

A. Phase 1: Contrastive Multimodal Pre-training

1) *Data pre-processing*: The workflow of data pre-processing is shown in Fig 2.

Firstly, we initiate the process by filtering the encrypted network traffic using the SSL/TLS protocol. This allows us

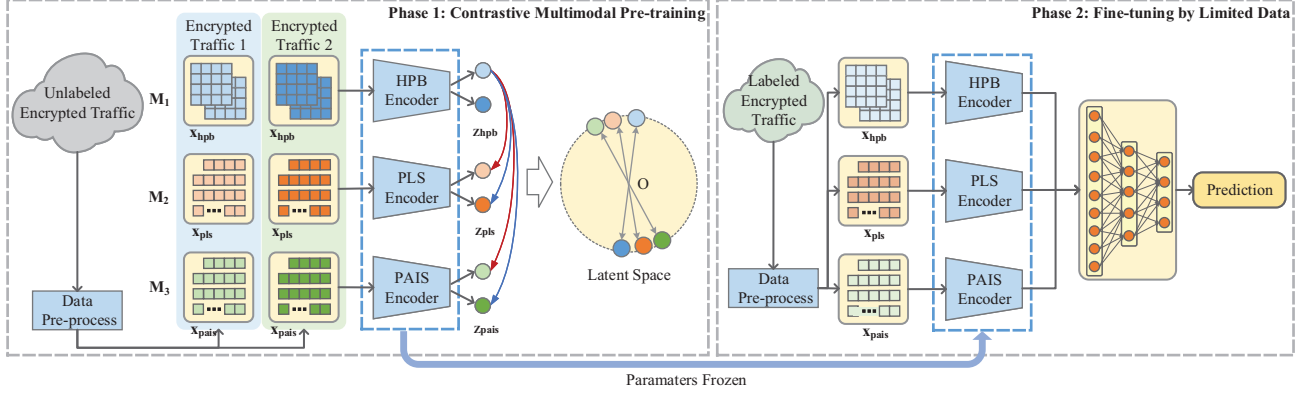


Fig. 1. Overview of CoMDet

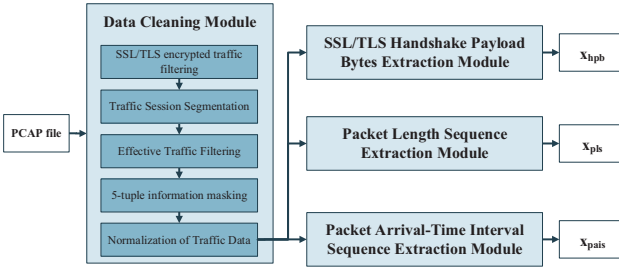
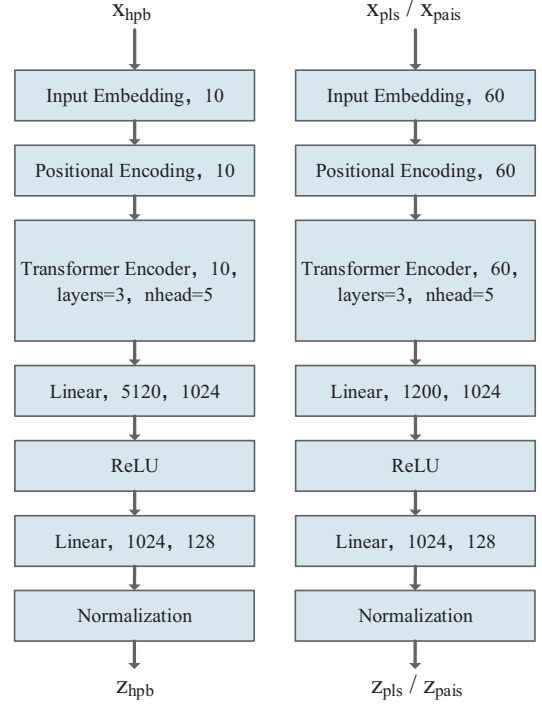


Fig. 2. The workflow of data pre-processing

to extract the traffic sessions based on the 5-tuple (*source IP, destination IP, source port, destination port, and transport layer protocol*). Secondly, we eliminate noises in data, such as failed sessions and retransmitted packets. This step ensures that we obtain valid encrypted network traffic samples. Thirdly, considering that the aforementioned 5-tuple information contains session characteristics that may impact algorithm performance, we choose to mask it with zeros prior to normalization. Finally, in accordance with the method outlined in MEMTD [9], we derive three distinct modalities respectively as SSL/TLS handshake payload bytes, packet length sequences, and packet arrival-time interval sequences.

2) *Transformer Encoder*: CoMDet uses 3 modalities for contrastive learning, which are the SSL/TLS handshake payload bytes $M_1 = \{x_{hpb}^1, x_{hpb}^2, \dots, x_{hpb}^N\}$, the packet length sequences $M_2 = \{x_{pls}^1, x_{pls}^2, \dots, x_{pls}^N\}$, and the packet arrival-time interval sequences $M_3 = \{x_{pais}^1, x_{pais}^2, \dots, x_{pais}^N\}$, where N is the pre-training dataset size. CoMDet uses three individual Transformer encoders for M_1 , M_2 , and M_3 . The encoders map the three modalities to the same latent space and optimize the parameters through inter-modal contrastive learning.

Fig 3 shows Transformer encoders are used for encoding the three modalities: the HPBEncoder, the PLSEncoder, and the PAISEncoder. Since the Transformer encoder abandons the recurring iterative operation for acceleration, a position



(a) HPBEncoder

(b) PLSEncoder/PAISEncoder

Fig. 3. Transformer encoders in CoMDet

embedding layer is needed to provide positional information for each input. The position embedding layer applies the sequence of positions to the input embeddings, which adds positional information about the element e_i , where i is the position in the sequence.

$$e_i = e_i + PE(i), \quad (1)$$

where, $PE(i)$ is the position embedding at position i , and the

value of $PE(i)$ at dimension j is represented as $PE(i, j)$. The calculation for embedding dimensions of d_{model} is as follows:

$$PE(i, 2k) = \sin(i/10000^{2j/d_{model}}). \quad (2)$$

$$PE(i, 2k+1) = \cos(i/10000^{2j/d_{model}}). \quad (3)$$

The output of the position embedding layer serves as the input to the Transformer encoder layer. The Transformer encoder layer consists of three individual Transformer encoders. Each encoder utilizes a multi-head attention mechanism with 5 heads, which effectively captures comprehensive information about the input sequence through self-attention throughout the encoding process. Subsequently, the output is projected onto various modalities through a series of fully connected layers. Finally, the normalization layer standardizes the feature vectors. After the encoding process, HPBEncoder, PLSEncoder, and PAISEncoder each yield the corresponding modality code representation: z_{hpb} , z_{pls} , and z_{pais} .

3) *Inter-modal contrastive learning*: The inter-modal contrastive learning process of the three modalities is detailed in Algorithm 1.

The objective function of the algorithm is as follows:

$$\begin{aligned} Loss_C(M_1, M_2, M_3) = & Loss(M_1, M_2) \\ & + Loss(M_2, M_3) \\ & + Loss(M_1, M_3), \end{aligned} \quad (4)$$

where $Loss(M_1, M_2)$ can be calculated as follows:

$$Loss(M_1, M_2) = Loss_{contrast}^{M_1, M_2} + Loss_{contrast}^{M_2, M_1}, \quad (5)$$

the calculation of $Loss_{contrast}^{M_1, M_2}$ can be expressed in the following:

$$Loss_{contrast}^{M_1, M_2} = -\frac{1}{k+1} \cdot \sum_{i=1}^{k+1} \left\{ \mathbb{E}_{\{m_1^i, m_2^i, \dots, m_2^{k+1}\}} \left[\log \frac{h_\theta(\{m_1^i, m_2^i\})}{\sum_{j=1}^{k+1} (h_\theta(\{m_1^i, m_2^j\}))} \right] \right\}. \quad (6)$$

f_{θ_1} and f_{θ_2} are the Transformer encoders for M_1 and M_2 respectively, where θ_1 and θ_2 are the parameters in Transformer encoders. The discriminant function h_θ in equation (6) can be represented as:

$$h_\theta(\{m_1, m_2\}) = \exp\left(\frac{f_{\theta_1}(m_1) \cdot f_{\theta_2}(m_2)}{\|f_{\theta_1}(m_1)\| \cdot \|f_{\theta_2}(m_2)\|}\right). \quad (7)$$

The calculation of $Loss(M_2, M_3)$ and $Loss(M_1, M_3)$ is similar to the above equations, we will not repeat it.

Algorithm 1: Inter-modal contrastive learning algorithm of CoMDet.

Input: Three modalities

$$M_1 = \{m_1^1, m_1^2, \dots, m_1^N\}, M_2 = \{m_2^1, m_2^2, \dots, m_2^N\}, M_3 = \{m_3^1, m_3^2, \dots, m_3^N\}.$$

Output: Multimodal Transformer encoders $f_{\theta_1}, f_{\theta_2}, f_{\theta_3}$.

Require: Initializing Transformer encoders $f_{\theta_1}, f_{\theta_2}, f_{\theta_3}$, Epoch E , Batch Size BS .

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1  $BN \leftarrow N/BS$ .
2 for  $e \in \{1, 2, \dots, E\}$  do
3   for  $k \in \{1, 2, \dots, BN\}$  do
4      $\{s_1^1, s_1^2, \dots, s_1^{BS}\} \leftarrow \{m_1^{(k-1)BS+1}, m_1^{(k-1)BS+2}, \dots, m_1^{(k-1)BS+BS}\}$ .
5      $\{s_2^1, s_2^2, \dots, s_2^{BS}\} \leftarrow \{m_2^{(k-1)BS+1}, m_2^{(k-1)BS+2}, \dots, m_2^{(k-1)BS+BS}\}$ .
6      $\{s_3^1, s_3^2, \dots, s_3^{BS}\} \leftarrow \{m_3^{(k-1)BS+1}, m_3^{(k-1)BS+2}, \dots, m_3^{(k-1)BS+BS}\}$ .
7      $S_1 \leftarrow \{s_1^1, s_1^2, \dots, s_1^{BS}\}$ .
8      $S_2 \leftarrow \{s_2^1, s_2^2, \dots, s_2^{BS}\}$ .
9      $S_3 \leftarrow \{s_3^1, s_3^2, \dots, s_3^{BS}\}$ .
10     $Loss \leftarrow 0$ .
11    for  $(i, j) \in \{(1, 2), (2, 1), (2, 3), (3, 2), (1, 3), (3, 1)\}$  do
12       $loss \leftarrow 0$ .
13      for  $p \in \{1, 2, \dots, BS\}$  do
14         $loss \leftarrow loss - \log \frac{h_\theta(\{s_i^p, s_j^p\})}{\sum_{q=1}^{BN} (h_\theta(\{s_i^p, s_j^q\}))}$ ,
15         $h_\theta(\{x, y\}) = \exp\left(\frac{f_{\theta_i}(x) \cdot f_{\theta_j}(y)}{\|f_{\theta_i}(x)\| \cdot \|f_{\theta_j}(y)\|}\right)$ .
16      end
17       $Loss_{contrast}^{S_i, S_j} \leftarrow \frac{1}{BS} loss$ .
18       $Loss \leftarrow Loss + Loss_{contrast}^{S_i, S_j}$ .
19    end
20    Adjust  $f_{\theta_1}, f_{\theta_2}, f_{\theta_3}$  by  $Loss$ .
21  end
22 end
23 return  $f_{\theta_1}, f_{\theta_2}, f_{\theta_3}$ .

```

Increasing the number of modalities learned through contrastive learning is more effective in developing an encoding method that maximizes modality inter-information. Hence, we use an encrypted traffic dataset containing three modalities. The objective function $Loss_C$ takes into account the mutual information that exists simultaneously in all three modalities. Therefore, CoMDet employs all modalities within the encrypted traffic for contrastive learning and optimizes the encoder for all three modalities concurrently. This method maximizes the mutual information that exists simultaneously in the data from all three modalities, as opposed to optimizing encoders for individual modalities in pairs. By computing the objective function simultaneously for all three modalities in equation (4), CoMDet achieves the goal of maximizing the mutual information among the data from all three modalities.

B. Phase 2: Fine-tuning by limited data

Unlike existing fine-tuning methods [12, 13], CoMDet incorporates a novel approach by utilizing limited data that

does not exist in the pre-training dataset, thereby introducing new categories of threats to simulate real-world scenarios as shown in Phase 2 of Fig 1. The model maps modalities to z_{hpb} , z_{pls} , and z_{pais} through the pre-trained HPBEncoder, PLSEncoder, and PAISEncoder. Then, the concatenated modalities representation $[z_{hpb}, z_{pls}, z_{pais}]$ is fed into a fully connected layer, which incorporates the mutual information among the encrypted traffic modalities in order to classify the encrypted traffic. It produces a prediction vector and offers a probability distribution for the predicted targeted category of malicious encrypted traffic.

During the fine-tuning phase, the cross-entropy loss function is employed. Meanwhile, the parameters pertaining to HPBEncoder, PLSEncoder, and PAISEncoder remain frozen, allowing solely the adjustment of fully connected layer parameters through the back-propagation algorithm. The fine-tuning time of CoMDet is significantly reduced due to the limited dataset size and the relatively fewer parameters in the Transformer encoders awaiting refinement. Consequently, the model can swiftly acquire the ability to classify new encrypted traffic generated by emerging instances of malicious encryption.

IV. EXPERIMENTS

In this section, we conduct experiments to evaluate CoMDet. We commence by providing a description of the dataset employed for evaluation, the baseline methods considered, and the metrics used to assess the outcomes. Subsequently, we undertake an analysis of CoMDet's overall performance, and conduct a detailed comparison between CoMDet and baseline methods, elucidating the underlying reasons behind these comparisons. Ultimately, we delve into the implications arising from alterations in the dataset size and the impact of malicious categories.

A. Experiment Setting

1) *Dataset*: We construct our dataset based on the Stratosphere Laboratory Datasets [20]. We remove the attack traffic with obvious patterns (such as IPs) and select 8 malicious categories including Bunitu, Caphaw, Dridex, HTBot, Miuref, Neris, TrickBot, and Zbot. The dataset statistic is shown in Table I.

To simulate the real-world scenario, we select 4 categories of malicious data and half of the benign data as the pre-training dataset in Phase 1. As for Phase 2, we construct a fine-tuning dataset with size n ($n \in \{10, 20, 40, 80\}$) consisting of the remaining 4 categories and benign encrypted traffic. The remaining data from these categories are used to evaluate the performance of the detection model as the test set. Based on the aforementioned dataset construction instructions, we can select 4 out of 8 categories of malicious encrypted traffic. This selection process generates a total of K dataset options to thoroughly assess the true effectiveness of the CoMDet, K is calculated as $C_8^4 = 70$ in our setting.

TABLE I
DATASET STATISTIC

Category	Pcap numbers	Encrypted session numbers
Bunitu	4	5,259
Caphaw	1	2,057
Dridex	3	111
HTBot	6	18,813
Miuref	3	2,499
Neris	3	4,365
TrickBot	4	522
Zbot	6	7,398
Benign	17	24,885
SUM	47	65,909

Each repetition involves various dataset construction selections, so we take the average value of the evaluation index across all repetitions. By employing this experimental setup, we aimed to minimize any discrepancies between the experiment and real-world scenarios. Furthermore, it allowed us to evaluate the overall performance of the experiments when dealing with diverse forms of malicious encrypted traffic.

2) *Baselines*: We use four encrypted malicious traffic detection methods as baselines:

Semi-supervised learning methods:

- 1D-CNN [13]: Rezaei *et al.* propose a semi-supervised learning method to train an encrypted traffic classification model using packet arrival-time interval sequences and packet sequences. This method first pre-trains a 1D-CNN model on an unlabeled encrypted traffic dataset. Then, they fine-tune the model with a few labeled samples to classify encrypted traffic effectively.
- FC-Net [12]: FC-Net learns a pair of feature maps for classification from a pair of encrypted traffic samples based on meta-learning, then compares the obtained feature maps and finally determines whether the pair of samples belong to the same category. This method can complete detection based on a small amount of encrypted malicious traffic samples.

Supervised learning methods:

- MEMTD [9]: MEMTD achieves the best performance among supervised deep learning methods in encrypted malicious traffic detection tasks. It adopts an intermediate fusion mechanism to learn the dependencies among modalities.
- FS-NET [21] FS-Net utilizes packet length sequences training with a deep learning model incorporating Bi-GRU (Bidirectional Gate Recurrent Unit). It effectively classifies traffic based on the flow features of encrypted data.

For supervised learning methods, we train pre-trained models by labeled data and fine-tune by uniform limited data.

3) *Metrics*: We use the $precision_{avg}$, $recall_{avg}$, ACC_{avg} and $macro-F1_{avg}$ to evaluate the detection performance,

which denotes the average of *precision*, *recall*, *ACC* and *macro-F1*, respectively. Metrics *precision*, *recall*, *ACC* and *macro-F1* are calculated as follows:

$$precision = \frac{TP}{TP + FP}, \quad (8)$$

$$recall = \frac{TP}{TP + FN}, \quad (9)$$

$$ACC = \frac{TP + TN}{TP + TN + FP + FN}, \quad (10)$$

$$macro-F1 = \frac{1}{C} \sum_{i=1}^C 2 \times \frac{recall \times precision}{recall + precision}. \quad (11)$$

True Positive (*TP*) means malicious samples are correctly identified as malicious. True Negative (*TN*) means benign samples are correctly identified as benign. False Positive (*FP*) means benign samples are incorrectly identified as malicious. False Negative (*FN*) means malicious samples are incorrectly identified as benign. *C* means encrypted traffic category numbers, $C = 5$ (4 malicious traffic categories and benign traffic) in our experiment.

B. Comparison with semi-supervised learning methods

Table II summarizes the detection performance and the improvements of CoMDet over the existing semi-supervised learning methods training with limited labeled data. In general, CoMDet achieves *macro-F1_{avg}* ranging between 64.7% and 86.56% and *ACC_{avg}* ranging between 72.56% and 92.44%. We can observe that CoMDet achieves an *ACC_{avg}* of nearly 90% using only 40 samples in each category. We also find that increasing the dataset size can effectively improve the detection performance of CoMDet, but this improvement slows down as the dataset size increases. It can be seen from Table II, the *ACC_{avg}* of CoMDet improved by 8.70% when the number of samples increased from 10 to 20 in each category. But the *ACC_{avg}* improvement becomes only 4.12% when the number of samples increases from 40 to 80. This indicates that for most categories of malicious traffic, distinguishable encrypted traffic samples can be captured with 20 to 40 samples, and additional encrypted traffic capture provides relatively limited classification information.

The 1D-CNN method only uses packet arrival-time interval sequences and packet sequences to detect encrypted malicious traffic, which performs worse than the FC-Net which uses encrypted traffic payloads as input. This illustrates that encrypted traffic payloads are more suitable than flow statistical features in training with limited data. FC-Net learns a pair of feature maps from a pair of encrypted traffic samples based on meta-learning. However, FC-Net falls short in conducting a comprehensive fusion analysis of diverse features, resulting in an inferior performance to CoMDet. CoMDet outperforms others with training on limited data since learning the mutual information among multiple modalities during the contrastive multimodal pre-training phase.

C. Comparison with supervised learning methods

The comparison of the CoMDet, FS-Net, and MEMTD is shown in Fig 4. Semi-supervised CoMDet gets comparable performance with FS-Net but is weaker than the MEMTD. However, as the number of samples used during fine-tuning increases, the performance gap between CoMDet and MEMTD gradually decreases. The *ACC_{avg}* and *macro-F1_{avg}* indicators of the CoMDet and MEMTD differ by 2.33% and 3.86% respectively when only the dataset size in each category becomes 40. The difference in these two indicators decreased to 1.24% and 2.57% when the dataset size increased to 80. This demonstrates that inter-modal contrastive learning can effectively learn features of encrypted traffic representations without requiring classification label guidance in the pre-training phase. This ability can solve the inconvenience caused by the difficulty in annotating datasets of encrypted traffic. Due to FS-Net utilizing packet length sequences without other modalities to train a deep learning model, it shows constraints when working with small dataset sizes.

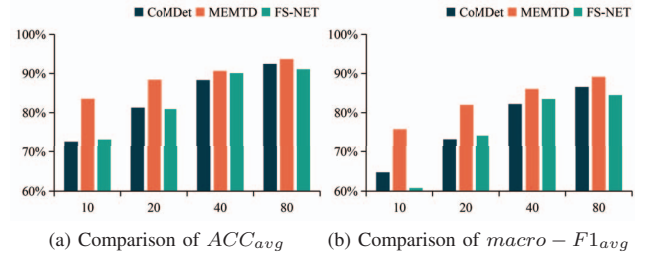


Fig. 4. Comparison with supervised learning methods with dataset sizes in each category

D. Encrypted malicious traffic categories analysis

Fig 5 illustrates the impact of various categories of encrypted malicious traffic on the detection performance of CoMDet. It is observed that CoMDet demonstrates effective classification ability for most categories of encrypted malicious traffic. However, its effectiveness in detecting encrypted malicious traffic from TrickBot and Dridex categories is comparatively weaker. Considering the dataset statistic provided in Table I regarding the encrypted malicious traffic data, it can be inferred that these two categories of encrypted malicious activities generate a smaller volume of encrypted traffic during their operation. Additionally, these two categories of traffic show lower similarity, which further limits CoMDet's detection performance when dealing with small samples of encrypted malicious traffic.

The addition of more samples does not lead to a significant improvement in detection performance when the dataset size in each category exceeds 40. This finding aligns with the observations presented in Table II, suggesting that fine-tuning CoMDet with limited data can achieve favorable detection performance. On the other hand, it necessitates an exponential

TABLE II
RESULTS OF SEMI-SUPERVISED LEARNING METHODS TRAINING WITH SMALL DATASET SIZES

Dataset size in each category	Method	$precision_{avg}$	$recall_{avg}$	ACC_{avg}	$macro-F1_{avg}$
10	1D-CNN [13]	0.5736	0.6783	0.6308	0.5967
	FC-Net [12]	0.5978	0.8028	0.6873	0.6276
	CoMDet	0.6321	0.8194	0.7256	0.6470
20	1D-CNN [13]	0.6243	0.7182	0.6923	0.6298
	FC-Net [12]	0.6438	0.8398	0.7586	0.6879
	CoMDet	0.7028	0.8801	0.8126	0.7315
40	1D-CNN [13]	0.6534	0.7823	0.7128	0.6876
	FC-Net [12]	0.7223	0.8876	0.7889	0.7321
	CoMDet	0.7867	0.9271	0.8832	0.8216
80	1D-CNN [13]	0.6926	0.8237	0.8098	0.7265
	FC-Net [12]	0.7911	0.9278	0.8370	0.7786
	CoMDet	0.8332	0.9536	0.9244	0.8656

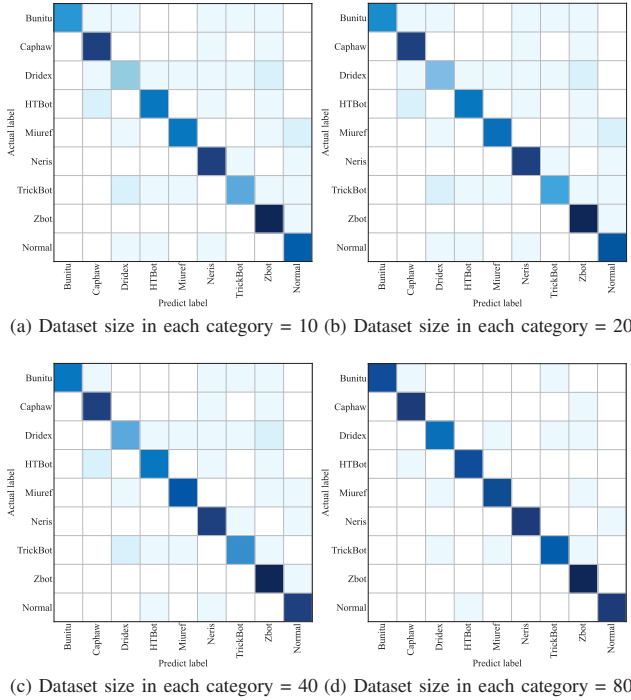


Fig. 5. Normalized confusion matrixes of CoMDet with different dataset sizes in each category

increase in the number of encrypted malicious traffic samples to achieve further enhancement in detection performance.

V. CONCLUSION

This paper presents CoMDet, which is a contrastive multimodal pre-training approach to encrypted malicious traffic detection. CoMDet is capable to learn multimodal feature representations by three individual Transformer encoders. It utilizes unlabeled data during the pre-training phase and fine-tuning with limited labeled data. CoMDet introduces

an inter-modal contrastive learning method to learn feature representations by maximizing mutual information among all modalities. The experimental results demonstrate that CoMDet surpasses existing semi-supervised learning methods and achieves comparable performance with existing supervised learning methods. Notably, we observe that as the dataset size increases in the fine-tuning phase, the detection performance of CoMDet gradually approaches the performance of supervised learning methods. Moreover, we conduct an in-depth analysis of CoMDet's performance across various categories of malicious traffic. In the future, we plan to investigate CoMDet's detection capability in real applications.

VI. ACKNOWLEDGEMENT

This paper is supported by GuangDong Basic and Applied Basic Research Foundation 2022B1515120072.

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